MIL-L-10470D(MU)
24 OCTOBER 1963
SUPERSEDING
MIL-L-10470C (ORD)
25 JANUARY 1962

MILITARY SPECIFICATION

LIGHTS, INSTRUMENT: FIRE CONTROL EQUIPMENT

1. SCOPE

1.1 This specification covers the general requirements for instrument lights used with fire control equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

J-C-98 — Cable and Wire, Insulated; Methods of Sampling and Testing.

MILITARY

MIL-F-13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel: Packaging of.

MIL-I-45208 — Inspection Requirements, General Specification for.

MIL-C-55040 — Cable, Special Purpose, Electrical (Retractile).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials.** The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 **Fabrication.** The instrument light shall be manufactured in accordance with the drawings applicable to model instrument light specified in the contract or order. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926.

(a) Order of precedence.

(b) Dimensions and tolerances.

FSC 1290

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:05Z
Check the source to verify that this is the current version before use.

(c) Inorganic protective surface finishes.

(d) Electrical and electronic assemblies.

(e) Part identification and marking.

(f) Workmanship.

3.4 Environmental.

3.4.1 *Operating temperatures.* The instrument light shall meet all the performance requirements of this specification while exposed to, and thermally stabilized at, ambient temperatures of plus 150 and minus 40 degrees Fahrenheit.

3.5 Methyl methacrylate parts. In instrument lights containing methyl methacrylate rods, the surface of the rods, unless otherwise specified, shall be highly polished. The rods shall be clear, transparent and without noticeable color. When sandblasting of sections is specified, care shall be taken to protect polished surfaces.

3.6 Electrical.

3.6.1 *Continuity.* The instrument light shall demonstrate continuity in accordance with the applicable electrical schematics pertaining thereto.

3.6.2 *Insulation resistance.* All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current applied.

3.7 Performance.

3.7.1 *Flicker.* There shall be no visible flicker when the indicator lamp is slightly jarred.

3.7.2 *Switch.* The snap switch shall have positive off and on action.

3.7.3 *Rheostat.* The wiper arm of the rheostat shall maintain positive contact with the windings of the rheostat for any setting of the rheostat control.

3.7.4 *Retractile cable.* When the instrument light contains a retractable lead cable, the cable shall have met all requirements of Specification MIL-C-55040.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL-STD-105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:05Z
Check the source to verify that this is the current version before use.

level and sampling plans specified in table I titled "Classification of Defects." Examinations and Tests for Packaging, Packing and Marking shall be in accordance with Specification MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE I. *Classification of defects*

Use Level II, of table I with Sampling Plan table II–A of Standard MIL–STD–105.

Critical: None

Major	AQL 1.5% defective	Requirement	Test procedure
101	Methyl methacrylate parts (when required).	3.5	Visual
102	Continuity	3.6.1	4.5.2
103	Insulation resistance	3.6.2	4.5.3
104	Flicker	3.7.1	4.5.4
105	Switch	3.7.2	4.5.5
106	Rheostat	3.7.3	4.5.6
107	Retractile	3.7.4	Specification MIL–C–55040.

4.2.3 *Disposition of nonconforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL–STD–105.

4.3 Control tests.

4.3.1 *General.* Three instrument lights shall be selected at random from each 100 produced or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedure*
Material	3.1	Visual Drawings & Specifications.
Fabrication	3.2	Visual — Drawing
General Specification	3.3	Specification MIL–F–13926.

4.3.2 *Operating temperatures.* Three instrument lights shall be selected at random from each 50 produced, or from each months production, whichever occurs first, to determine operability in conformance with 3.4.1 and 4.5.1.

4.3.3 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment," and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction,

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:05Z
Check the source to verify that this is the current version before use.

materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 *Operating temperatures.* The instrument lights shall be exposed to the temperatures indicated in 3.4.1, utilizing testing equipment in accordance with the "Test facilities" requirements of Specification MIL-F-13926. This equipment shall be capable of maintaining these temperatures until thermal stabilization is reached. The temperatures shall be changed gradually to avoid thermal shock. While stabilized at each extreme temperature condition the units under test shall be subjected to and conform to the requirements specified in 3.4.1.

4.5.2 *Continuity.* A continuity check of the instrument lights shall be made using standard measuring devices to determine conformance with the requirements specified in 3.6.1.

4.5.3 *Insulation resistance.* This test shall be performed in accordance with the test Procedure 6031 of Specification J-C-98 using the voltage and current specified in this specification, and shall conform to the requirement of 3.6.2.

4.5.4 *Flicker.* The instrument light shall be securely held in any position and the indicator lamp shall be lighted from a suitable source of ac or dc power. The instrument light then shall be jarred by striking to visually examine the lamp, for flicker in conformance to the requirements specified in 3.7.1.

4.5.5 *Switch.* The snap switch for the instrument light shall be manually operated to the "OFF" position and to the "ON" position. The switch shall be visually checked for conformance to the requirements specified in 3.7.2.

4.5.6 *Rheostat.* The rheostat control shall be slowly rotated from one extreme position to the other while observing illumination. Illumination shall be constant and with uniform change in brightness as the knob is rotated to insure compliance with 3.7.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The instrument light shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-P-14232 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. Instrument lights are intended to illuminate dials, scales, reticles, and other graduated parts of fire control instruments.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging, and packing.

(c) Applicable packaging data sheet number (see 5.1).

(d) Applicable stock number.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. 1290-A144Y

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:05Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
U.S. ARMY MUNITIONS COMMAND
ATTN: SMUFA-5500
PHILA., PA. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:05Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

C-6270

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:05Z
Check the source to verify that this is the current version before use.